Tracking of Optically Propelled Nanoparticles in Hollow-Core Fiber Using an Event-based Camera

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***Abstract*—Event-based tracking of 150-nm-diameter silica nanoparticles propelled within an anti-resonant hollow-core fiber is demonstrated. Through the integration of defocused particle imaging with a two-step trajectory tracking framework, continuous and high-precision nanoparticle velocimetry is achieved.**

Keywords—Event-based camera; hollow-core fiber; optical tweezers; nanoparticle tracking

I. Introduction

Hollow-core fiber (HCF) offers a unique platform for the optical manipulation and sensing of nanoparticles within confined light fields [1]. In HCF, nanoparticles can be levitated at the core center via optical gradient forces and propelled along the fiber by optical forces [1, 2]. In this context, accurate tracking and velocity measurement are essential for controlled particle guidance and for achieving high-resolution sensing functionalities [3, 4, 5]. Doppler velocimetry, which relies on monitoring the back-scattered light from the particles, has previously been demonstrated for high-resolution velocity tracking [2, 3, 6]. However, this approach faces significant challenges in resolving the absolute particle position, as it requires integrating the instantaneous velocity over time, which leads to accumulated errors that grow with increasing propagation distance [4].

Unlike frame-based cameras, which redundantly sample static scenes while failing to capture fast motion adequately, event-based camera (EBC) leverages a biomimetic neuromorphic design inspired by retinal motion sensing [7]. In EBC, each pixel independently detects brightness changes and asynchronously generates signed events only when a predefined threshold is exceeded, producing a sparse stream of contrast events instead of redundant intensity frames [8], [9]. This event-driven architecture offers microsecond temporal resolution, a high dynamic range while maintaining low data rates and power consumption [10, 11]. In the context of optical tweezes, Ren et al. demonstrated high-bandwidth tracking of levitated microparticles [10] and, more recently, real-time control of particle arrays [11] using EBC.

Precise motion tracking of nanoparticles in HCF using an EBC remains challenging, as the scattering signal is strongly distorted by the fiber cladding and the inherent scattered light intensity from the nanoparticle is weak [12, 13]. In this work, we report high-resolution tracking and velocity measurement of a 150-nm-diameter SiO_2 particle in HCF using an EBC. By integrating defocused particle imaging with a two-step trajectory tracking framework, we achieve continuous and high-precision nanoparticle velocimetry. A comparison with Doppler velocimetry is also presented.

II. Experimental Setup

The experimental setup is schematically illustrated in Fig. 1. An anti-resonant HCF with a core diameter of 33 μm is employed for particle trapping and manipulation. A 1064-nm laser beam is coupled into the fundamental core mode to optically trap and propel 150-nm-diameter SiO_2 particles along the fiber. The imaging branch comprises an EBC positioned beside the fiber to record the light laterally scattered by the particle. A triangular aperture is additionally introduced in front of the EBC, such that the effective scattering signature manifests as a recognizable bright triangular pattern on the image plane.

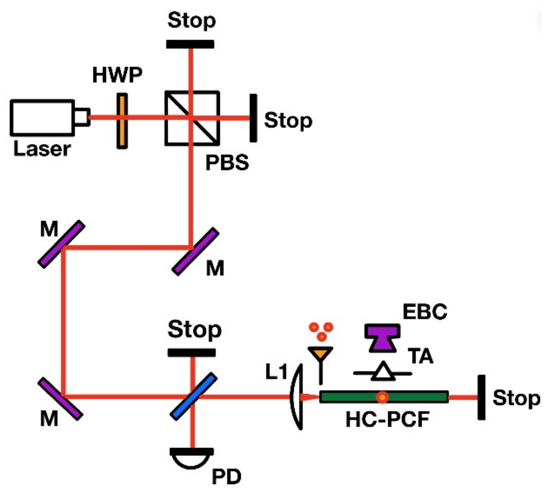

Fig. 1. Experimental setup. HWP, half wave plate; PBS, polarizing beam splitter; PD, photodiode; EBC, event-based camera; TA, triangular aperture.

III. Experimental Results

The event streams captured by the EBC are accumulated and exported as grayscale video with slow-motion factor $\gamma = 10$. The mapping between total frame number N_{tot} and real time t_r is given by:

$$t_r = \frac{N_{tot}}{f_s \cdot \gamma} \tag{1}$$

where f_s is the video frame rate. To mitigate the influence of scattering pattern distortion induced by the fiber cladding, the imaging plane is intentionally defocused to produce a more uniform scattering spot at the EBC. As shown in Fig. 2, the focused spot (Fig. 2(a)) is highly non-uniform and exhibits strong fluctuations, whereas the defocused spot is less sensitive to distortion while retaining the aperture-shaped geometry for subsequent motion tracking.

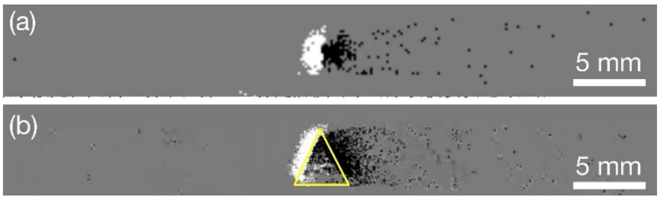

Fig. 2. The captured (a) focused and (b) defocused imaging of lateral scattered light from the 150-nm-diameter SiO_2 particle. The yellow-marked area indicates the triangular template determined by the algorithm.

Triangular apertures offer superior feature distinguishability over circular or square apertures of equal area, as demonstrated by Kang et al. in single-frame coherent diffraction imaging [14]. In our setup, this geometry projects a distinct triangular scattering signature onto the image plane (Fig. 2(b)), enabling reliable template-based tracking.

In the signal processing pipeline, the triangular aperture projects a distinct illumination pattern onto the image plane, providing a geometrically constrained feature that facilitates centroid extraction (see the yellow-marked area in Fig. 2(b)). Conventional brightness-weighted centroid estimation based on intensity thresholding is susceptible to background-induced false detections, which introduce abrupt displacement errors into the particle trajectory. To address this issue, a two-step tracking framework is introduced. In the first step, In the first step, the initial 200 frames are used to calibrate the nominal particle velocity v_c and construct a canonical triangular template that propagates at this velocity (Fig. 2(b)). In the second step, particle localization is performed within the triangular template search window centered at the predicted position. When template matching fails, the predicted template moving at the nominal velocity v_c is adopted to maintain trajectory continuity. The obtained trajectory is then used to extract the particle velocity by projecting the positions onto the fiber axis and subsequently differentiating with respect to time. Finally, a moving-average filter is applied to suppress outliers and attenuate high-frequency noise.

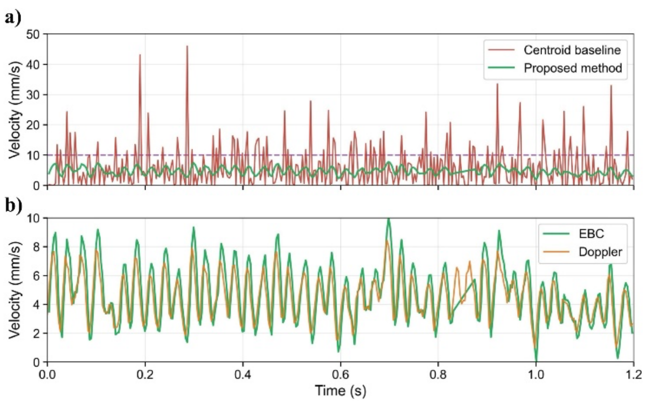

Fig. 3. (a) Resolved particle velocity by the conventional centroid baseline tracking (red) and proposed method (green). (b) Resolved particle velocity by the EBC and Doppler velocimetry. The dashed lines mark the regime where valid template matching is achieved.

To validate the proposed method, the EBC video is exported at 10× slow motion. Fig. 3(a) compares the velocity extraction results obtained by the conventional centroid-based baseline tracking method (red) and the proposed method (green) on the same slow-motion video (with a total of 452 frames, N_{tot} = 452). The dashed lines indicate the regime where valid template matching is achieved. As observed, the conventional baseline method yields only 84.1% valid frames. In contrast, the proposed method achieves 100% valid frames and produces a continuous trajectory, confirming that the proposed algorithm significantly enhances both detection reliability and velocity stability under defocused nanoparticle scattering conditions.

To further validate the proposed method, the EBC-based particle velocimetry is compared with Doppler velocimetry (Fig. 3(b)). In the Doppler measurement configuration, the backscattered light from the particle is converted into a spectrogram, from which the ridge frequency is extracted to serve as a global velocity reference [3]. The velocity profiles obtained from the EBC and Doppler velocimetry exhibit consistent periodic variations over time, thereby confirming the feasibility of the proposed technique.

IV. Conclusion

We present an EBC-based frame-resolved velocimetry approach for 150-nm-diameter SiO_2 nanoparticles propelled in an anti-resonant HCF. By integrating defocused triangular-aperture imaging with a two-step tracking framework, robust particle detection and accurate velocity estimation are achieved. Comparison with Doppler velocimetry further validates the proposed technique. This method enables rapid tracking of nanoparticle trajectories and velocities within HCF, with promising applications in targeted particle delivery, nanoparticle metrology, and fiber-optic sensing.

References

[1] R. Wang et al., "Optical trapping of mesoscale particles and atoms in hollow-core optical fibers: Principle and applications," Light Sci. Appl., vol. 14, no. 1, p. 146, Mar. 2025.

[2] M. K. Garbos, T. G. Euser, O. A. Schmidt, S. Unterkofler, and P. St. J. Russell, "Doppler velocimetry on microparticles trapped and propelled by laser light in liquid-filled photonic crystal fiber," Opt. Lett., vol. 36, no. 11, pp. 2020–2022, Jun. 2011.

[3] R. Wang et al., "Non-Markovian Doppler velocimetry of optically propelled microparticles in hollow-core photonic crystal fiber," ACS Photonics, vol. 11, no. 4, pp. 1533–1539, Apr. 2024.

[4] D. S. Bykov, O. A. Schmidt, T. G. Euser, and P. St. J. Russell, "Flying particle sensors in hollow-core photonic crystal fibre," Nat. Photonics, vol. 9, pp. 461–465, 2015.

[5] A. Sharma, S. Xie, R. Zeltner, and P. St. J. Russell, "On-the-fly particle metrology in hollow-core photonic crystal fibre," Opt. Express, vol. 27, pp. 34496–34504, 2019.

[6] O. A. Schmidt, M. K. Garbos, T. G. Euser, and P. St. J. Russell, "Metrology of laser-guided particles in air-filled hollow-core photonic crystal fiber," Opt. Lett., vol. 37, pp. 91–93, 2012.

[7] A. Vanarse, A. Osseiran, and A. Rassau, "A review of current neuromorphic approaches for vision, auditory, and olfactory sensors," Front. Neurosci., vol. 10, p. 115, 2016.

[8] G. Gallego, T. Delbrück, G. Orchard, C. Bartolozzi, B. Taba, A. Censi, S. Leutenegger, A. J. Davison, J. Conradt, K. Daniilidis, and D. Scaramuzza, "Event-based vision: A survey," IEEE Trans. Pattern Anal. Mach. Intell., vol. 44, pp. 154–180, 2022.

[9] P. Lichtsteiner, C. Posch, and T. Delbruck, "A 128×128 120 dB 15 μs latency asynchronous vision sensor," IEEE J. Solid-State Circuits, vol. 43, no. 2, pp. 566–576, Feb. 2008.

[10] Y. Ren, E. Benedetto, H. Borrill, Y. Savchuk, M. Message, K. O'Flynn, M. Rashid, and J. Millen, "Event-based imaging of levitated microparticles," Appl. Phys. Lett., vol. 121, p. 113506, 2022.

[11] Y. Ren, B. Siegel, R. Yin, Q. Wu, J. Pritchett, M. Rashid, and J. Millen, "Neuromorphic detection and cooling of microparticles in arrays," Nat. Commun., vol. 16, p. 10658, 2025.

[12] D. Grass, J. Fesel, S. G. Hofer, N. Kiesel, and M. Aspelmeyer, "Optical trapping and control of nanoparticles inside evacuated hollow core photonic crystal fibers," Appl. Phys. Lett., vol. 108, no. 22, p. 221103, May 2016.

[13] S. Xie, A. Sharma, M. N. Romodina, N. Y. Joly, and P. St. J. Russell, "Tumbling and anomalous alignment of optically levitated anisotropic microparticles in chiral hollow-core photonic crystal fiber," Sci. Adv., vol. 7, no. 28, p. eabf6053, Jul. 2021.

[14] J. Kang, S. Takazawa, N. Ishiguro, and Y. Takahashi, "Single-frame coherent diffraction imaging of extended objects using triangular aperture," Opt. Express, vol. 29, no. 2, pp. 1441–1452, Jan. 2021.